Tsz Fei Wang

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Education

University of Waterloo | Honours Computer Science (Co-op) — Sept 2026 – May 2031 (expected)

Skills & Technologies

Languages & Frameworks: Python, Java, C++, C, TypeScript, JavaScript, SQL, FastAPI, Node.js, PyTorch, Pandas
Tools: Linux, Docker, Git, GitHub Actions, REST APIs, Elasticsearch, Firebase, CosmosDB, Cloudflare Workers, Caddy, Postman

Experience

Century Frontier Asset Management — Jul 2026 – Aug 2026
Quantitative Research Intern — *Shenzhen, CN*

- Built a point-in-time backtesting pipeline in **Python** and **pandas** that eliminated lookahead bias from the firm's LLM-driven equity research across **104 tickers**.
- Benchmarked **9 LLM architectures** on stock-preference and inter-stock-relationship prediction, showing that a model's stated confidence was a poor predictor of its accuracy.

Krembil Research Institute, World Community Grid — Mar 2026 – Jun 2026
Software Engineer Intern — *Toronto, ON*

- Built the full-stack data access portal for Catrin, serving **~4 million** TF-TG associations across **33,000 genes** to researchers.
- Reconciled the schemas of **9 independent data sources** so the portal can query them as one dataset.

Mackenzie Cybersecurity Team — Jun 2025 – Jun 2026
Team Captain, President — *Toronto, ON*

- Captained the team to **First in Canada** at CyberTitan, Canada's largest cyberdefense competition, hardening Windows, Linux and Cisco systems under time pressure.
- Created and taught a cybersecurity curriculum reaching **60+ students** across the school.

Non-Trivial Research Fellowship — Mar 2025 – Aug 2025
Machine Learning Research Fellow — *Remote*

- Trained and evaluated a **PyTorch** audio deepfake classifier in **Python** on a Kaggle dataset of real and AI-synthesized speech.
- Researched AI voice cloning as an emerging security threat and proposed a detection-based defence in a written research proposal.

Projects

Physical Studio | *Python, React, Three.js, Docker* | ***Winner**, Hack the Valley 11* — *Oct 2026*

- Engineered the **Python** MCP server for a prompt-to-hardware build guide generator, letting an LLM design from a library of measured parts and check its own design for collisions and clearance.
- Built the **React Three Fiber** viewer that animates each assembly step and deployed the app with **Docker Compose** on Vultr.

huddle | *Node.js, Cloudflare Workers, MCP* — *Oct 2026 – Present*

- Designed a coordination layer in **JavaScript** that lets multiple Claude Code agents on separate machines edit one repository at once, using per-file locks that make other checkouts read-only and stream edits to them live.
- Implemented a dependency-free **Node.js** relay server on **Cloudflare Workers** that brokers locks, chat and live file edits between agents.
- Added git history sync between checkouts as chunked bundles, branch-scoped locks, and an end-to-end test suite that runs real Claude sessions against the live server.

Submerge | *GLSL, QML, Python* — *Oct 2026 – Present*

- Shipped an animated KDE Plasma theme in **GLSL** and **QML** downloaded **~1,290 times** on the KDE Store, with a shader water simulation and a live system HUD.
- Resolved a user-reported NVIDIA/Wayland rendering failure by porting the shaders to GLSL ES, and added power-aware frame pacing to save battery.

Awards & Honors

Gold & Silver Medalist, IT Systems Networking, Skills Ontario (Regional & Provincial) — Mar 2026 – May 2026
National Champion x2, 1st in Canada, Cybersecurity, $1,000 scholarship — May 2025, Mar 2026
National Finalist, 2nd in Canada, Cybersecurity, $1,600 scholarship — Mar 2025